

MARKETING CAMPAIGN INTELLIGENCE

Marketing Performance Analysis & Business Insights Report

1. Problem Statement

Marketing campaign data can be difficult to use for decision-making when each brand or campaign report is reviewed separately. Looking at individual reports makes it harder to compare brands on the same basis and answer practical questions such as which campaign type performs better, which channel combinations give stronger returns, and which customer groups should be explored further.

The project addresses this problem by creating one standardized BI model for three brands—Nykaa, Purpille, and Tira. The model brings campaign, cost, revenue, conversion, customer, targeting, and time-related information together so performance can be compared in one dashboard.

The aim is not just to show numbers, but to make the analysis easier to understand and turn the findings into practical recommendations for future marketing decisions.

2. Project Objective

The main objective was to build a simple but useful marketing intelligence solution that can support campaign performance comparison and budget-related decisions.

- Standardize the data for Nykaa, Purpille, and Tira into one master table.

- Validate the data before using it for analysis.

- Compare brands, campaign types, channel combinations, customer segments, and target audiences.

- Analyze changes in ROI, revenue, conversions, and conversion rate over time.

- Build an interactive Power BI dashboard for management-level and detailed analysis.

- Convert the analysis into clear business insights and recommendations.

3. Project Requirements

The project requirements were both technical and business-focused.

Requirement	Purpose
Data validation	Check duplicates, missing values, date ranges, and unusual minimum values.
Common master table	Combine all three brands so the same analysis can be applied consistently.
Performance analysis	Compare revenue, acquisition cost, ROI, leads, and conversions.
Campaign analysis	Compare campaign types and channel combinations.
Customer analysis	Identify higher-performing customer segments and target audiences.
Time analysis	Understand how ROI changes month by month.

BI dashboard	Provide interactive views for management and detailed analysis.
Business recommendations	Connect the analysis to possible marketing actions.

4. Dataset

The project uses a synthetic Multi-Brand Marketing Campaign Performance dataset. Each brand contains 55,555 records, giving a combined total of 166,665 records.

Brand	Records
Nykaa	55,555
Purpille	55,555
Tira	55,555
Total	166,665

The dashboard covers the period from July 2024 to June 2025.

Important fields include Campaign_ID, Campaign_Type, Target_Audience, Duration, Channel_Used, Impressions, Clicks, Leads, Conversions, Revenue, Acquisition_Cost, ROI, Language, Engagement_Score, Customer_Segment, and Date.

5. Technical Approach

5.1 Data Validation in MySQL

The first step was to check whether the three source tables were suitable for analysis. SQL queries were used to check record counts, duplicate Campaign IDs, NULL values, date ranges, and minimum values of important numerical columns.

The project analysis found no duplicate Campaign IDs and no important missing values. Negative ROI values were retained because a negative ROI can represent a loss-making campaign. Removing those rows would make the analysis less complete.

5.2 Creating the Master Table

A common table named marketing_campaigns was created. The three brand tables were inserted into it and a Brand column was added. This created one consistent dataset for SQL analysis and Power BI.

5.3 SQL Analysis

SQL was then used to compare overall brand performance, channel combinations, campaign types, customer segments, target audiences, Purpille's College Students segment, monthly performance, conversion rate, cost per conversion, and ROAS.

For channel analysis, complete channel combinations were kept together. For example, 'WhatsApp, Email, Facebook' was treated as one campaign combination instead of splitting it into three separate channels. This avoids double-counting campaign-level revenue and conversions.

5.4 Power BI Data Model

The master data was brought into Power BI and a DAX DateTable was created using the minimum and maximum campaign dates. The DateTable contains Date, Year, Month, Month Number, Year Month, and Year Month Sort. A one-to-many relationship was created from DateTable[Date] to marketing_campaigns[Date].

5.5 DAX Measures

Measures were created for Total Revenue, Total Acquisition Cost, Total Impressions, Total Clicks, Total Leads, Total Conversions, Average ROI, Conversion Rate, Cost Per Conversion, and Total

Campaigns. A Best Campaign measure was also used to identify the campaign type with the highest average ROI.

5.6 Dashboard Development

The final Power BI report contains four pages: Home, Overview, Campaign Performance, and Customer VS Targeting Insights. The pages are designed to move from a simple project introduction to executive KPIs and then to more detailed campaign and customer analysis.

6. Power BI Dashboard

The screenshots below show the actual dashboard pages used in the project.

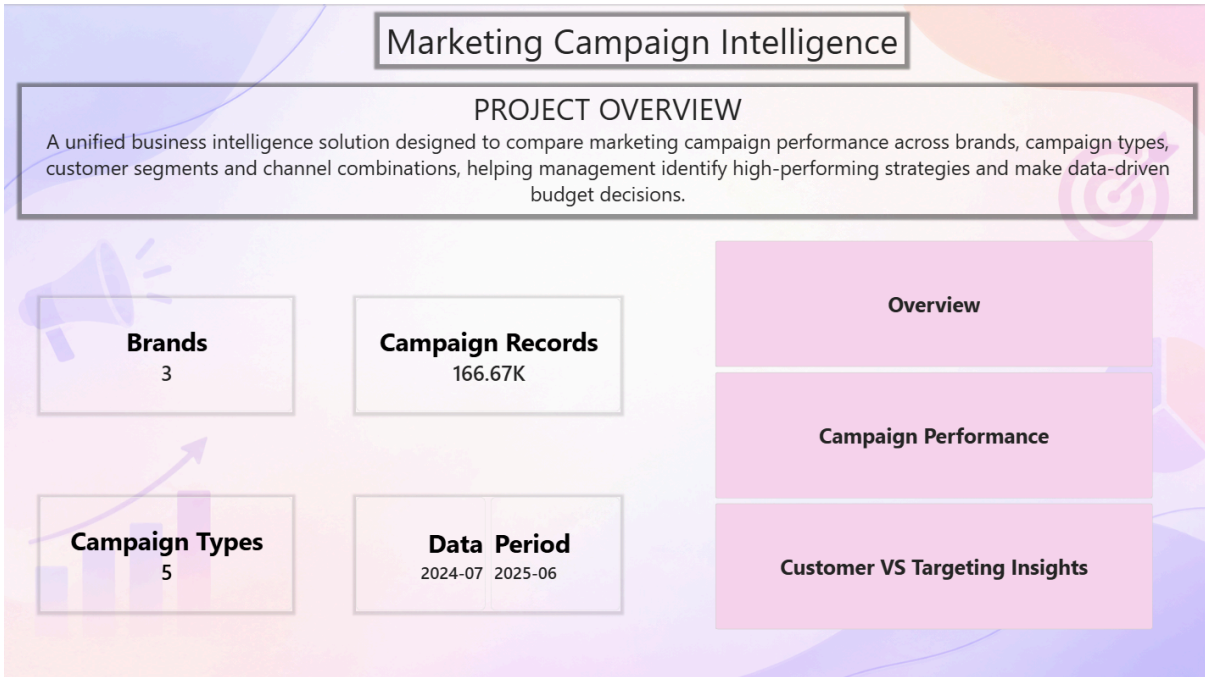


Figure 1. Home page

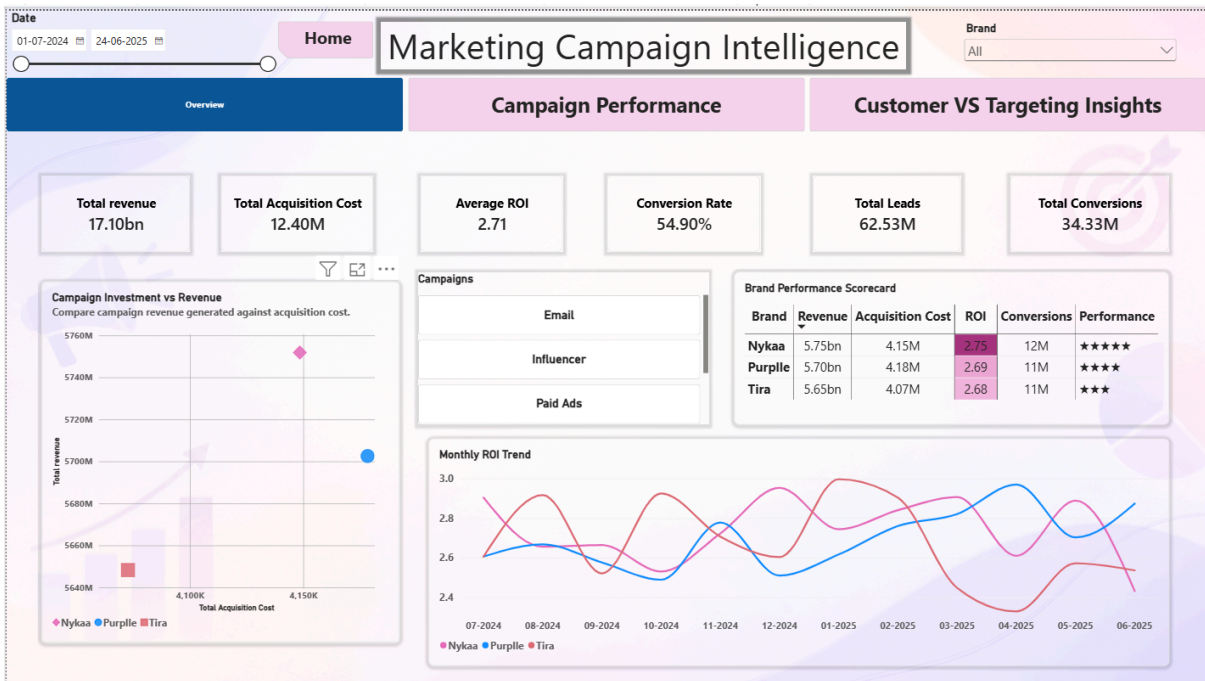


Figure 2. Overview page

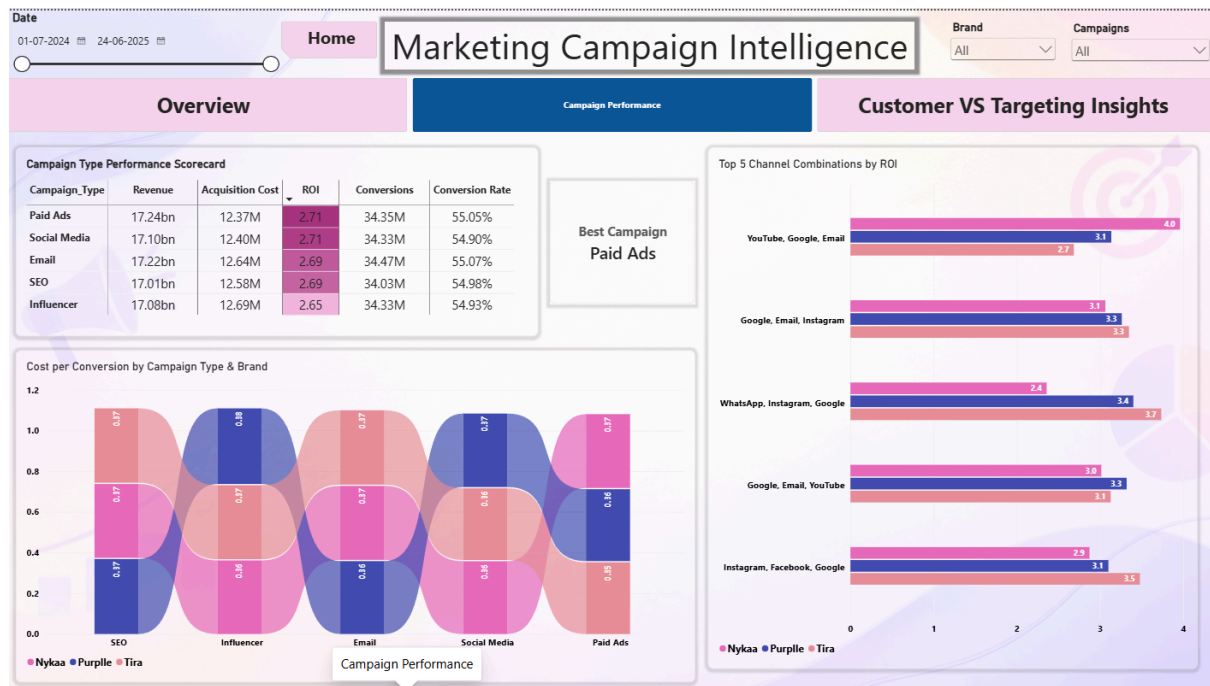


Figure 3. Campaign Performance page

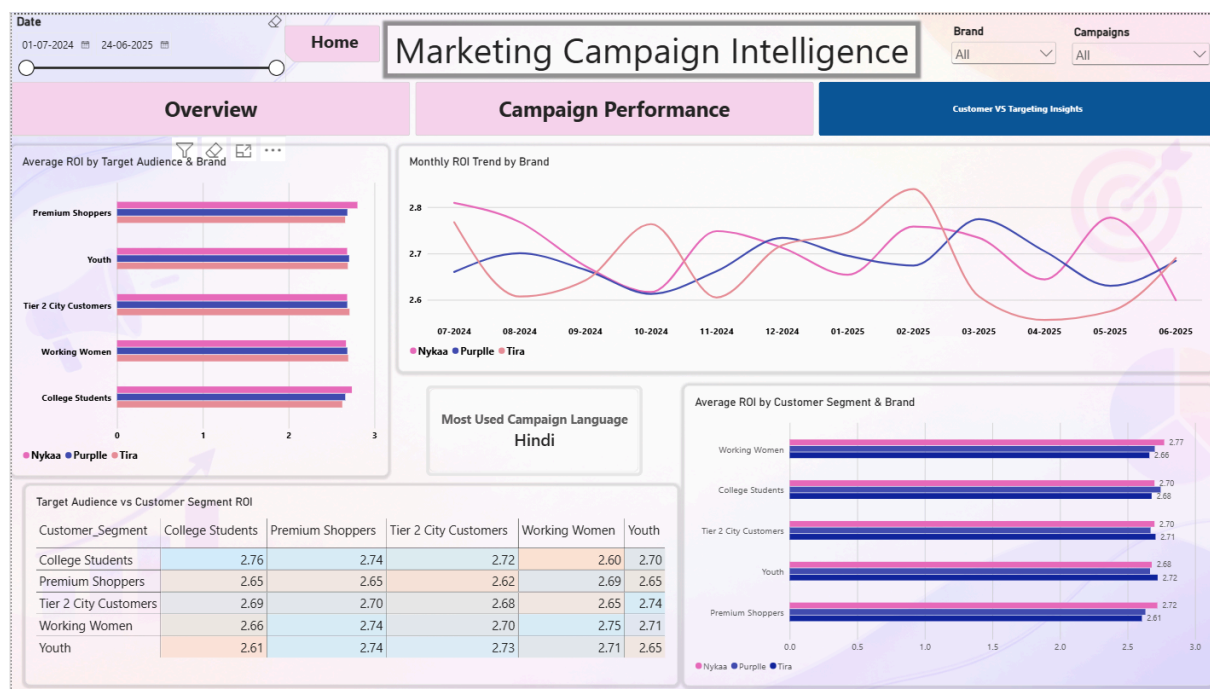


Figure 4. Customer VS Targeting Insights page

7. Dashboard-Level Observations

The Overview page gives an overall view of the selected period. In the displayed view, total revenue is 17.10bn, acquisition cost is 12.40M, average ROI is 2.71, conversion rate is 54.90%, total leads are 62.53M, and total conversions are 34.33M.

The brand scorecard shows Nykaa with 5.75bn revenue and 2.75 average ROI, followed by Purple at 5.70bn and 2.69 ROI, and Tira at 5.65bn and 2.68 ROI. The star ratings are relative ROI rankings in the dashboard and are not real-world ratings.

The Campaign Performance page shows Paid Ads as the Best Campaign in the displayed overall scorecard. Paid Ads and Social Media both show 2.71 ROI, while Email and SEO are around 2.69 and Influencer is 2.65.

The Customer VS Targeting page shows that performance varies by customer segment and target audience. It also includes monthly ROI trends and a target-audience-versus-customer-segment matrix.

8. Key Insights

8.1 Nykaa has the strongest overall performance

Nykaa records the highest revenue and average ROI among the three brands in the displayed dashboard: 5.75bn revenue and 2.75 average ROI. Purplle follows at 2.69 and Tira at 2.68.

The difference is relatively small, so the result should be treated as a performance lead rather than a large gap.

Possible explanation: the analysis shows that campaign type, channel combination, customer segment, and target audience performance differ by brand. Therefore, Nykaa's stronger overall result may be related to its mix of strategies and audiences. The available analysis does not prove that one specific factor caused the higher ROI.

8.2 Paid Ads is the strongest overall campaign type in the dashboard

Paid Ads is shown as the Best Campaign and has an average ROI of 2.71. Social Media also has 2.71, while Email and SEO are 2.69 and Influencer is 2.65.

The conversion rates are very close across campaign types, mostly around 55%. This means the small ROI differences are not explained by a large conversion-rate difference. They are more likely related to the relationship between acquisition cost and revenue in this synthetic dataset.

This is a useful point for budget discussions: a campaign type should not be judged only by how many conversions it produces; cost and revenue also matter.

8.3 Channel combinations perform differently by brand

The SQL analysis found different high-performing channel combinations for the three brands. Examples include Nykaa's WhatsApp, Email, Facebook combination at around 4.23 ROI, Purplle's Facebook, Email, YouTube at around 3.52, and Tira's WhatsApp, Instagram, Google at around 3.73.

Possible explanation: different brands may have different responses to combinations of communication and acquisition channels. However, the dataset does not provide enough evidence to say why customers responded better to one combination. These results should therefore be treated as testing opportunities rather than proven causes.

8.4 The strongest customer segment is different for each brand

Nykaa's highest-performing customer segment is Working Women, Purplle's is College Students, and Tira's is Youth.

This is important because it suggests that the same customer strategy should not automatically be applied to all three brands. Each brand can use its own strongest segment as a starting point for campaign experiments.

8.5 Target audience performance also changes by brand

Nykaa shows Premium Shoppers as its strongest target audience, Purplle shows Youth, and Tira shows Tier 2 City Customers.

The result supports a more brand-specific approach to targeting rather than assuming that one audience will work equally well across all brands.

8.6 Purple shows a potential targeting-strategy gap

College Students is Purple's highest-performing customer segment overall. However, within that segment, the target-audience analysis shows higher ROI for Tier 2 City Customers and Premium Shoppers than for campaigns directly labelled College Students.

The important point is not that Purple is targeting the wrong audience. Instead, the data suggests a potential targeting-strategy gap. Purple could test different targeting approaches, messages, or audience definitions to understand how to reach the College Students segment more effectively.

8.7 Monthly ROI changes over time

The monthly ROI charts show that the brands do not follow the same pattern throughout July 2024 to June 2025. Each brand has stronger and weaker months.

Possible explanation: campaign performance can change with the mix of campaigns running during a month. The available data does not include enough information about promotions, seasonality, or external events to identify a specific cause.

For this reason, monthly trend analysis is useful when reviewing performance instead of relying only on a full-period average.

8.8 Conversion rate is relatively stable

The displayed conversion rate is about 54.90%, and campaign-type conversion rates are generally close to 55%. This means conversion efficiency is fairly similar across the main campaign types in this dataset.

Because the conversion-rate differences are small, decisions should also consider ROI, revenue, and acquisition cost rather than focusing on conversion rate alone.

9. Business Recommendations

- Use Paid Ads and Social Media as priority candidates for controlled budget experiments because they are among the strongest overall campaign types by ROI.

- Evaluate channel combinations at brand level instead of applying one common channel mix to all three brands.

- Use the strongest customer segment for each brand as a starting point for targeted campaign testing.

- For Purple, run controlled tests inside the College Students segment using different target-audience definitions and campaign messaging. The goal should be to find a more effective way to reach the segment, not to stop targeting it.

- Review monthly ROI before making budget changes. A campaign type or brand with a good overall average can still have weaker individual periods.

- Use cost per conversion together with ROI and revenue when comparing campaign efficiency.

10. Important Observations and Limitations

- The dataset is synthetic. The values are suitable for demonstrating analysis skills but should not be presented as real industry benchmarks.

- The dashboard's star ratings are relative ROI rankings between selected brands, not customer ratings.

11. Conclusion

The project successfully brings campaign data for Nykaa, Purple, and Tira into one standardized analytical model. SQL was used to validate and analyze the data, while Power BI was used to build an interactive dashboard for executive, campaign, customer, and targeting analysis.

The main takeaway is that marketing performance is not the same across brands, campaign types, channel combinations, or customer groups. Nykaa leads overall in the displayed results, Paid Ads is the strongest overall campaign type, and the best customer and target audiences differ by brand.

The Purple analysis also shows why looking at two dimensions together can be more useful than looking at one metric alone. College Students is the strongest customer segment, but some other target audiences perform better within that segment. This creates a practical area for further testing.

Overall, the project demonstrates the complete workflow from data validation and SQL analysis to BI modelling, dashboard creation, business insights, and recommendations.